
Honors Differential Equations

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MA 231 Section A1

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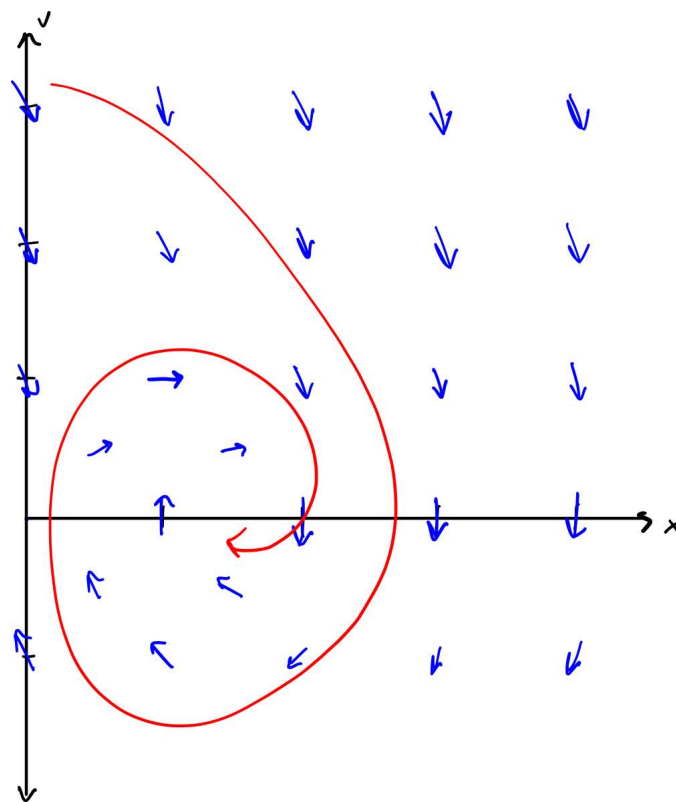
Section 2.2 Problem 19

(a,b,e)

$$\frac{dx}{dt} = v.$$

$$\frac{dv}{dt} = -2v + 3x - x^3.$$

(a)



(b) Set $\frac{dx}{dt}, \frac{d^2x}{dt^2} = 0$. It follows that $v, \frac{dv}{dt} = 0$. We thus have

$$x^3 - 3x = 0.$$

Note that I flipped the negative signs here. It follows that

$$x(x^2 - 3) = 0 \implies x \in \{0, \pm\sqrt{3}\}.$$

(e) The solutions appear to spiral around the equilibrium points $\pm\sqrt{3}$ and 0. There appears to be some level of overall tendency towards the equilibrium point $+\sqrt{3}$.

solns curl into the equilib it kinda goes down on positive thne up into

Section 2.4 Problem 13

(a) Clearly,

$$y = y_0 e^{-3t}.$$

We then have

$$\frac{dx}{dt} = 2x - 8y_0^2 e^{-6t}.$$

We use the method of integrating factors.

$$\mu(t) = e^{\int -2 dt} = e^{-2t}.$$

$$x(t) = -e^{2t} \int e^{-2t} 8y_0^2 e^{-6t} dt = -e^{2t} \int 8y_0^2 e^{-8t} dt = y_0^2 e^{-6t} + x_0 e^{2t}.$$

Thus,

$$x(t) = x_0 e^{2t} + y_0^2 e^{-6t}, \quad y(t) = y_0 e^{-3t}.$$

(b) Let $\frac{dx}{dt}, \frac{dy}{dt} = 0$. We have $y = 0$ trivially. It follows that $2x = 0$, and thus $x = 0$. Therefore, $(0, 0)$ is the only equilibrium point.

(c) For $y_0 = 1$, we have

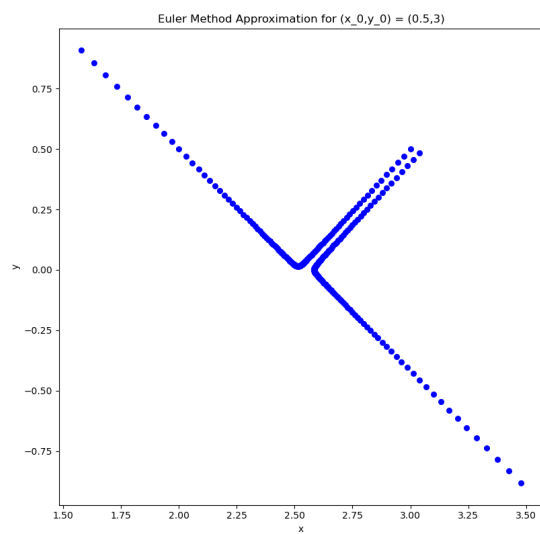
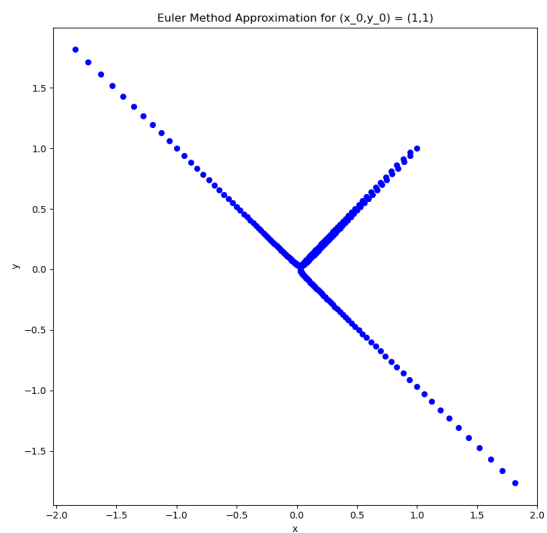
$$y(t) = e^{-3t}.$$

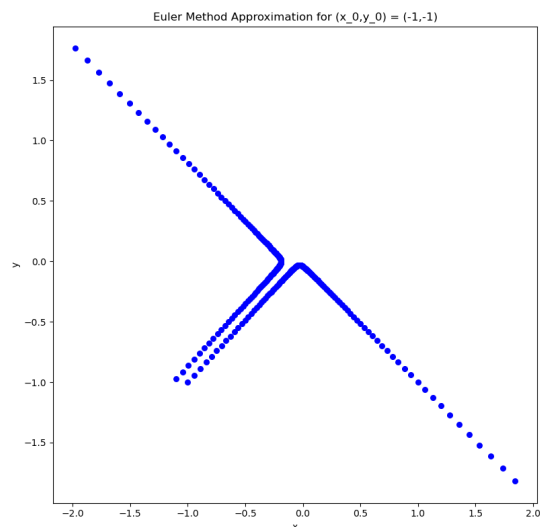
For $x_0 = 0$, we have

$$x(t) = e^{-6t}.$$

Note that we choose $t = 0$ since Picard's theorem applies in the xy -plane, so the choice of t is irrelevant.

(eulers)





Review 23

True. You can see the solution curve passing through $(0, 1)$ passes through $(0, 0)$. Thus, taking $(1, 0)$ to be the initial condition implies for some t , (x, y) will equal $(0, 0)$.

Review 24

False. Picard's theorem applies, and the point $(1/2, 0)$ is within the loop that goes from $(0, 0)$ to $(1, 0)$ and back. Thus, the graph must remain within this loop, and cannot tend to infinity.

Review 25

True. Since they lie on the same line, they are the same equation.

Section 3.1 Problem 6

$$\frac{d\mathbf{Y}}{dt} = \begin{bmatrix} 0 & 3 \\ -0.3 & 3\pi \end{bmatrix} \mathbf{Y}(t).$$

Section 3.1 Problem 17

(a) Define $v = \frac{dy}{dt}$. We have

$$\frac{dv}{dt} + pv + qy = 0, \quad \frac{dy}{dt} = v.$$

Define $\mathbf{Y}(t) = (y, v)$. We have

$$\frac{d\mathbf{Y}}{dt} = \begin{bmatrix} 0 & 1 \\ -q & -p \end{bmatrix} \mathbf{Y}(t).$$

If $q = 0$, we have

$$\frac{d\mathbf{Y}}{dt} = \begin{bmatrix} 0 & 1 \\ 0 & -p \end{bmatrix} \begin{bmatrix} y \\ v \end{bmatrix} = v \begin{bmatrix} 1 \\ -p \end{bmatrix}.$$

It follows that

$$\frac{dv}{dt} = -pv \implies v = e^{-pt}.$$

Additionally, We do not need to find y in terms of t , since if we substitute $\frac{d\mathbf{Y}}{dt} = \mathbf{0}$, we have

$$\mathbf{0} = v \begin{bmatrix} 1 \\ -p \end{bmatrix}.$$

Since the equation is no longer dependent on y , all values of y are equilibria. This can be validated by setting $\frac{d^2y}{dt^2}, \frac{dy}{dt} = 0$ in the original equation:

$$qy = 0.$$

Since $q = 0$, y can be anything.

(b) Let $p, q = 0$. We have

$$\mathbf{0} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y \\ v \end{bmatrix} \implies v = 0.$$

Since this equation does not depend on y still, all points are equilibria. We can verify this using the original equation:

$$\frac{d^2y}{dt^2} = 0.$$

Section 3.1 Problem 24

(a)

$$\begin{aligned} \frac{d\mathbf{Y}_1}{dt} &= \frac{d}{dt} \begin{bmatrix} 0 \\ e^t \end{bmatrix} = \begin{bmatrix} 0 \\ e^t \end{bmatrix}. \\ \frac{d\mathbf{Y}_1}{dt} &= \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ e^t \end{bmatrix} = \begin{bmatrix} 0 \\ e^t \end{bmatrix}. \\ \frac{d\mathbf{Y}_2}{dt} &= \frac{d}{dt} \begin{bmatrix} e^{2t} \\ e^{2t} \end{bmatrix} = \begin{bmatrix} 2e^{2t} \\ 2e^{2t} \end{bmatrix}. \\ \frac{d\mathbf{Y}_2}{dt} &= \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} e^{2t} \\ e^{2t} \end{bmatrix} = \begin{bmatrix} 2e^{2t} \\ e^{2t} + e^{2t} \end{bmatrix}. \end{aligned}$$

(b) Every solution is a linear combination of our basis solutions $\mathbf{Y}_1(t), \mathbf{Y}_2(t)$, since they are linearly independent. Thus,

$$\mathbf{Y}(0) = c_1 \mathbf{Y}_1(t) + c_2 \mathbf{Y}_2(t).$$

We have

$$\begin{bmatrix} -2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ c_1 \end{bmatrix} + \begin{bmatrix} c_2 \\ c_2 \end{bmatrix} \implies \begin{bmatrix} -2 \\ -1 \end{bmatrix} = \begin{bmatrix} c_2 \\ c_2 + c_1 \end{bmatrix}.$$

We have $c_2 = -2$, and thus $c_1 = 1$. Therefore,

$$\mathbf{Y}(t) = \mathbf{Y}_1(t) - 2\mathbf{Y}_2(t).$$

Supplimental 1

$r = 0.8$		$r = 2$		$r = 3.2$	
n	z_n	n	z_n	n	z_n
0	0.100000	0	0.100000	0	0.100000
1	0.072000	1	0.180000	1	0.288000
2	0.053453	2	0.295200	2	0.656179
3	0.040476	3	0.416114	3	0.721946
4	0.031071	4	0.485926	4	0.642368
5	0.024084	5	0.499604	5	0.735140
6	0.018803	6	0.500000	6	0.623069
7	0.014760	7	0.500000	7	0.751533
8	0.011634	8	0.500000	8	0.597540
9	0.009199	9	0.500000	9	0.769555
10	0.007291	10	0.500000	10	0.567488
11	0.005790	11	0.500000	11	0.785425
12	0.004605	12	0.500000	12	0.539304
13	0.003667	13	0.500000	13	0.795057
14	0.002923	14	0.500000	14	0.521413
15	0.002332	15	0.500000	15	0.798533

For $r = 0.8$, we notice that z_n appears to be asymptotically approaching 0. It appears as though 0 is an equilibrium point.

For $r = 2$, notice that z_n rapidly approaches 0.5 and stabilizes there. Clearly, 0.5 is an equilibrium point. In the continuous case, z_n would not actually reach 0.5, and it's possible it actually hasn't reached it and there are simply too many decimals for my code to handle. However, it's possible that since the system is discrete, it is allowed to reach and stabilize at the equilibrium point in finite time. It appears to approach it fairly linearly, as compared to asymptotically in the case of $r = 0.8$.

For $r = 3.2$, we notice that it rapidly approaches 0.65, but then slows down and begins to oscilate between 0.6ish and 0.7ish. It appears to be oscillating around an equilibrium point. The most interesting part is that the oscillations appear to be *increasing* in amplitude, as compared to a general continuous case, where they usually decrease. Notice how they start around 0.72 and 0.65, decrease slightly, then go up to 0.79 and 0.52, which is a significantly larger delta. This is unlike the behaviour of any of the previous solutions.

I was so curious about the case for $r = 3.2$ that I ran the code for larger values of n .

n	z_n
0	0.100000
1	0.288000
2	0.656179
3	0.721946
4	0.642368
5	0.735140
6	0.623069
7	0.751533
8	0.597540
9	0.769555
10	0.567488
11	0.785425
12	0.539304
13	0.795057
14	0.521413
15	0.798533
16	0.514810
17	0.799298
18	0.513346
19	0.799430
20	0.513093
21	0.799451
22	0.513052
23	0.799455
24	0.513046
25	0.799455
26	0.513045
27	0.799455
28	0.513045
29	0.799455
30	0.513045

It appears to stabilize at two different points; instead of approaching a single equilibria, it oscillates between two different points. I hope we learn more about this behavior, because this is very interesting.